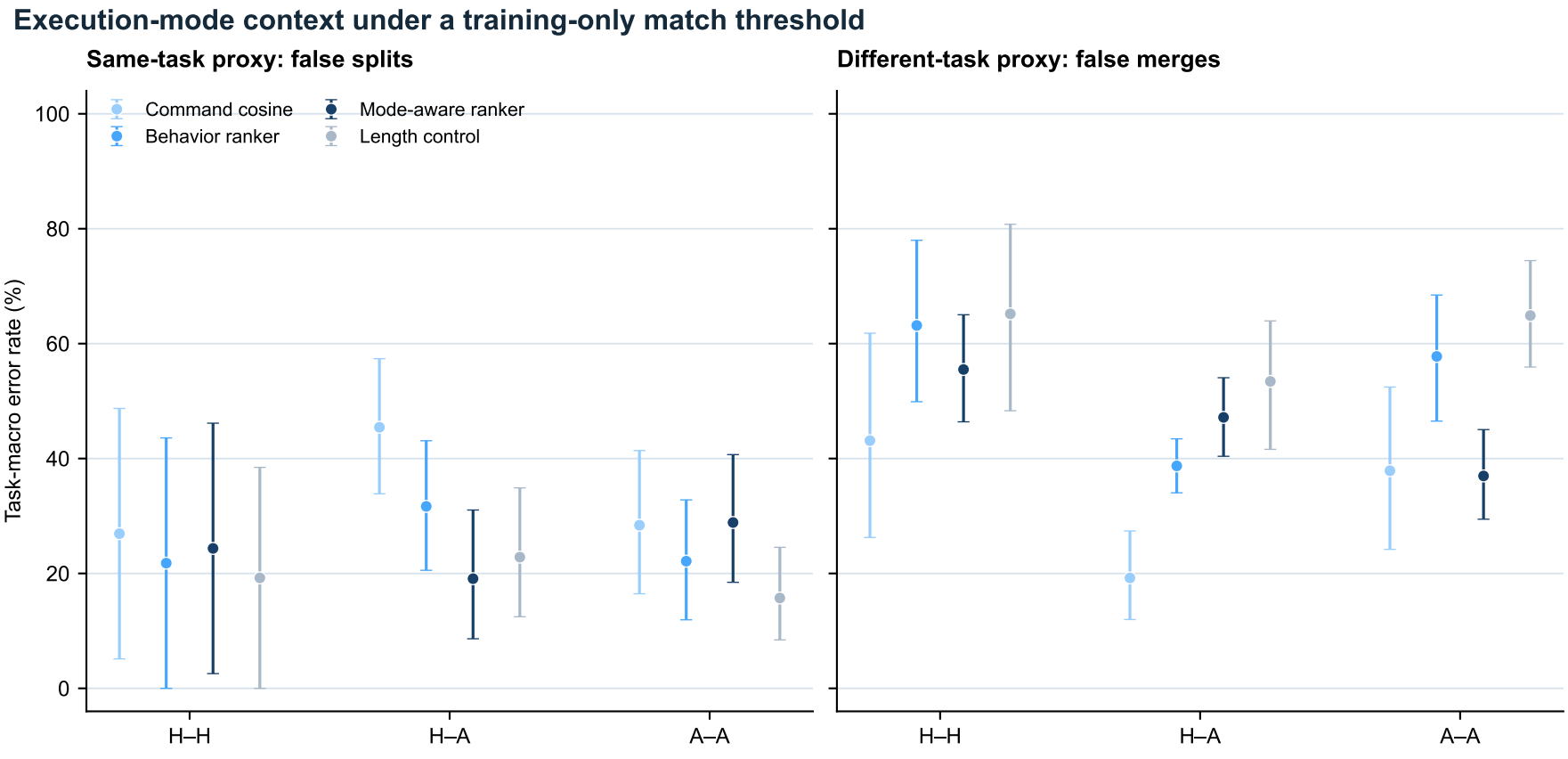
Execution-mode context under a training-only match threshold
Same-task proxy: false splits
Different-task proxy: false merges
Command cosine
Behavior ranker
Mode-aware ranker
Length control
Task-macro error rate (%)
100
80
60
40
20
0
H–H
H–A
A–A
H–H
H–A
A–A
Pairs are labeled by shared CTF task only; this is not attack-activity ground truth. Error bars: 95% task-cluster bootstrap CI.